# IPCON® PPS GF

## Glass Fiber Reinforced Polyphenylene sulfide Filament

**This is the filament's Technical Data Sheet (TDS). If you want the Material Safety Data Sheet (MSDS), please contact the customer service to download.**

### • Product Information

| Item | Data |
|---|---|
| Filament Diameter | 1.75 ± 0.05 mm |
| Net Filament Weight | 0.75 kg ± 0.5% |
| Filament Length | 230 m ± 2% |
| Spool Material | Cardboard (max heat resistance: 150 °C)<br>or high-temp PC (max heat resistance: 140 °C) |
| Spool Size | Outer Cardboard's diameter: 200 mm;<br>Inner roller's diameter: 110 mm |
| Recommendation for body protection during use | This material contains glass fiber, which may cause skin irritation. So it is recommended to wear gloves throughout the operation. |

### • Filament's Properties

| Physical Properties | Testing Methods | Data |
|---|---|---|
| Density | ISO 1183-3 | 1.36 g/cm³ |
| Shrinkage Rate* | Specially-defined | 0.21% - 0.42% |
| Surface Hardness | Shore hardness | 83D |
| Saturated Water Absorption Rate | 25 °C, 55% RH, room air | 0.05% |
| Melt Index | 340 °C, 2.16 kg | 22.3 g/10 min |
| Melting Temperature | DSC, 10 °C/min | 278 °C |
| Vicar Softening Temperature | ISO 306, GB/T 1633 | 258 °C |
| Heat Deflection Temperature | ISO 75, 0.45 MPa | 241 °C |
| Relative Thermal Index (RTI) Max Continuous Service Temperature | Actual working conditions simulation test | 190 - 210 °C<br>(Related to load, size, shape, wall thickness, etc., and the value after annealing is higher than that before annealing in general) |

| Mechanical Properties* | Testing Methods | Data* |
|---|---|---|
| Tensile Strength XY | ISO 527, GB/T 1040 | 72 MPa |
| Tensile Strength Z | ISO 527, GB/T 1040 | 42 MPa |
| Young's Modulus XY | ISO 527, GB/T 1040 | 5764 MPa |
| Young's Modulus Z | ISO 527, GB/T 1040 | 2581 MPa |
| Breaking Elongation Rate XY | ISO 527, GB/T 1040 | 1.6% |
| Breaking Elongation Rate Z | ISO 527, GB/T 1040 | 0.7% |

| | | |
|---|---|---|
| Flexural Strength XY | ISO 178, GB/T 9341 | 112 MPa |
| Flexural Strength Z | ISO 178, GB/T 9341 | 58 MPa |
| Flexural Modulus XY | ISO 178, GB/T 9341 | 4530 MPa |
| Flexural Modulus Z | ISO 178, GB/T 9341 | 2212 MPa |
| Impact Strength XY | ISO 179, GB/T 1043 | 31.7 kJ/m² |
| Impact Strength Z | ISO 179, GB/T 1043 | 10.6 kJ/m² |

*The shrinkage rate and mechanical properties were tested on specimens printed by IPCON using a Bambu H2S printer (nozzle temperature: 345 °C, bed temperature: 120 °C, printing speed: 100 mm/s) and IPCON PPS GF (Grey) filament. The parameters of specimens of mechanical properties are as following: infill density—100%, infill pattern—concentric; and those of the 2 models of shrinkage (2 crosses that are vertical in the X and Y directions with designed lengths of 100.00 and 150.00 mm respectively) are as following: wall loop—2 and 6, infill density—15% and 25%, infill pattern—grid. The data is for user reference and comparison only. When using it, users are responsible for the safety, legal compliance and product performance.

## • Main Features and Application Scenarios

1. High stiffness, high strength, insulated, easy to print (compatible with Bambu H2D, H2S, H2C, and QIDI Q2, Q1 Pro and other printers that have 350 °C nozzle temperature and positive chamber temperature, as well as AMS or CFS), top-grade flame retardancy (can meet UL94 V-0 @0.8 mm/1.6 mm/3.2 mm), ultra temperature resistance (long-term temperature resistance of 200 °C), superb water resistance and solvent resistance
2. Suitable for automotive engines and fluid systems, aerospace, chemical industry, medical disinfection equipment and electrical equipment and other application scenarios that require flame retardancy, insulation, heat resistance or water and solvent resistance.

## • Recommended Printing Settings

| | |
|---|---|
| Drying Settings Before Printing | 100 - 140 °C, 6 - 8 h (Blast drying oven) |
| Printing Temperature and Humidity | ≤ 120 °C, ≤ 20% RH<br>(Sealed with desiccant) |
| Storage Temperature and Humidity | ≤ 35 °C, ≤ 20% RH<br>(Sealed with desiccant) |
| Compatible Support Material | Itself or support filament for PPS |
| Compatible Printer Type | Enclosed-frame<br>(60 to 120 °C chamber temperature is recommended) |
| AMS or CFS | Compatible |
| Compatible Nozzle Material | Hardened steel, diamond, tungsten carbide, etc |
| Compatible Nozzle Size | 0.6 mm (recommended) / 0.4 mm / 0.8 mm |
| Compatible Plate Type | Smooth / Textured PEI Plate or other plate |
| Plate Surface Preparation | Apply liquid glue or PVP solid glue stick if the first layer bonding is not firm enough |
| Nozzle Temperature | 320 - 360 °C |
| Bed Temperature | 100 - 120 °C |
| Chamber Temperature | 60 - 120 °C |
| Fan Speed | 0% - 40% |
| Max Print Speed | 300 mm/s |

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